Tendon-driven anthropomorphic hand with actuators housed in the hand skeleton, single-spool antagonistic tendons, load-based adaptive grasping and two-channel EMG control, for robotic and prosthetic use

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- **OSHWA certification:** IT000021, 4 April 2025, https://certification.oshwa.org/it000021.html

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Abstract

An anthropomorphic hand, manufacturable by 3D printing, usable both as a robotic end effector and as an active prosthetic hand. Five rotary actuators are housed inside a single internal skeleton of the hand, to which the fingers, the palm part and the covers are attached. Each finger is moved by a flexion tendon and an extension tendon fixed to the same spool on one actuator and wound in opposite directions, so one actuator actively both closes and opens the finger. Tendons are routed through holes in the printed phalanges and anchored by self-tightening wraps. Ring and little fingers share one actuator through a crimped clamp and a sliding tendon loop, giving an adaptive differential closure. The thumb base is mounted directly on the output flange of a thumb rotation actuator. Servo position and load feedback are used for automatic tendon calibration at start-up and for grasps that close each finger until a load limit is reached. Grasp presets map a single 0–100% closure command to different finger trajectories. In the prosthetic version, two surface EMG sensors are processed by FFT; the signal intensity of each is compared with its own threshold to increase, decrease or hold the closure value.

Keywords

robotic hand; prosthetic hand; artificial hand; gripper; end effector; tendon-driven; cable-driven; antagonistic tendons; flexor tendon; extensor tendon; spool; capstan; underactuated; differential coupling; adaptive grasp; force-limited grasp; servo load feedback; tendon calibration; thumb opposition; 3D-printed hand; myoelectric prosthesis; grasp preset; EMG; electromyography; FFT; threshold control; grasp selection; back-drivable thumb

Background

Low-cost tendon-driven hands often place actuators in the forearm, which prevents use as a compact robot end effector, or use a single tendon per finger with elastic return, which gives weak and imprecise opening. Calibration of tendon length is usually manual and is

lost when tendons stretch. Fixed position commands either crush soft objects or fail to grasp irregular ones.

Description

1. Skeleton and actuators (Figs. 2, 3, 4)

A single internal skeleton part, covered by a separate palm part and by front and back covers, contains five slots, one per actuator: index flexion, middle flexion, ring-and-little flexion, thumb flexion and thumb rotation. In the reference implementation the actuators are serial bus servomotors (WaveShare ST3215) connected in a daisy chain, each fixed by four M2 screws. The upper fingers are inserted in slots at the top of the skeleton and fixed by one M3 bolt each. A tendons-barrier part is screwed to the rear of the skeleton to separate and guide the tendons towards the spools.

2. Thumb (Fig. 5)

The thumb base is bolted directly onto the output flange of the thumb rotation actuator, so that the actuator rotates the whole thumb about an axis that produces opposition. The thumb is bolted to the thumb base and is flexed by a tendon driven by the thumb flexion actuator located in the skeleton.

3. Fingers and tendon routing (Fig. 7)

Each finger is made of printed phalanges. A tendon enters from the rear of the skeleton, crosses to the front, and is threaded in sequence through holes in the proximal, medial and distal phalanges on the palmar side (flexion path). At the distal phalanx it passes to the dorsal side and back, and is wrapped twice around its own incoming segment; tension on the tendon tightens these wraps, anchoring it to the distal phalanx without knots or glue. The tendon then runs back through holes on the dorsal side of the phalanges (extension path), exits at the rear of the skeleton, passes over the tendons-barrier and reaches the spool. Tendons are threaded using a loop of thin nylon line used as a needle.

4. Single-spool antagonistic actuation (Fig. 6)

A spool is bolted to the output flange of each flexion actuator. The flexion tendon is wound about half a turn in one direction and locked under a bolt head in a hole of the spool; the extension tendon is wound about one turn in the opposite direction and locked under a second bolt. Rotation of the actuator in one direction winds the flexion tendon and unwinds the extension tendon, closing the finger; rotation in the opposite direction opens it. Both motions are actively driven by the same actuator, without return springs. During installation the firmware moves the actuators to a predefined tendon installation position; each tendon is pulled until the finger is at its maximum opening and then locked.

5. Ring–little differential coupling (Fig. 8)

Ring and little fingers each have their own tendon routed as in section 3. At the rear of the skeleton the ends of the two tendons are joined in one crimped clamp, with different free lengths between each finger exit hole and the clamp (e.g. 5 cm and 4.5 cm), which sets the relative closing sequence. A further tendon, connected to the spool of the shared actuator, ends in a small loop (about 3 mm diameter) that slides along the tendon segment between the two fingers. When one finger is blocked by the object, the loop slides and the other

finger continues closing, so the two fingers adapt to the object shape with a single actuator.

6. Electronics

Robotic version: a commercial ESP32 servo driver board. Prosthetic version: a custom board with an ESP32 module, a 12 V to 5 V switching regulator, and single-gate tri-state buffers (sn74lvc1g125 / sn74lvc1g126) converting the microcontroller UART into the half-duplex serial bus of the servomotors. Supply: 12 V DC.

7. Control method (Figs. 9, 10)

a. **Start-up tendon calibration.** For each flexion actuator the controller moves the finger towards opening in small increments, reading the actuator load at each step, until the load exceeds a threshold (finger fully extended against its tendons); it then does the same towards closing. The positions found are stored as the actual range of the finger, compensating tendon length and stretch without manual adjustment.

b. **Grasp presets.** A grasp type (power, small power, "monkey"/hook, pinch, relax, power tool) defines, for each degree of freedom, a trajectory as a function of a single closure value from 0 to 100%. The user or robot sends only the grasp type and the closure value.

c. **Force-limited adaptive closing.** On closing, each finger advances in increments while its load is read; each finger stops independently when its load reaches a per-finger limit or when its target position is reached. The hand therefore wraps around objects of different shape and softness with a limited force. A global torque limit is also set on each actuator.

d. **RAW mode.** The closure of each degree of freedom is set independently.

e. **Two-channel EMG control (prosthetic version).** Two surface EMG sensors are placed on two muscle groups, one assigned to opening and one to closing. In each control cycle the controller samples each sensor (in the reference firmware 512 samples at about 9.9 kHz), removes the DC component, applies a Hamming window and computes the FFT magnitude; the intensity is taken as the magnitude in a frequency bin around 200 Hz. Each channel has its own intensity threshold. Decision rule: - open-channel intensity above its threshold → decrease the closure value (open), regardless of the close channel; - open channel below threshold and close channel above its threshold → increase the closure value (close); - both below their thresholds → keep the closure value unchanged. The closure value is changed by a fixed step (5% in the reference firmware) per cycle and limited to 0–100%; it is applied to the selected grasp preset, so a single proportional value drives all fingers along the preset trajectories, with the force-limited closing of point c. Giving priority to the open channel makes releasing an object always possible.

f. **Automatic relax.** If the closure value stays at 0% for a set time (60 s in the reference firmware), the hand automatically switches to the relaxed grasp posture.

g. **Grasp selection by manual thumb positioning.** While the hand is open, the controller disables the torque of the thumb rotation actuator, so the user can rotate the thumb by hand with the other hand. The controller reads the actuator position and selects the grasp preset whose thumb rotation position is closest (within a tolerance), e.g. power, small power, hook, pinch or relax. Torque is enabled again when closing starts. The thumb thus works as a mechanical grasp selector, without extra buttons or sensors.

h. **Grasp switching by EMG pattern.** As an option, the sequence of recent EMG decisions (open / close / none) is stored in a short buffer while the hand is open; a double "open" pulse pattern cycles to the next grasp preset.

i. **Command protocol.** Plain-text commands over a serial link (Bluetooth in the reference implementation), framed as `#CMD##\r\n`: grasp names select the grasp, a number sets the closure, and `I<n>M<n>R<n>T<n>TR<n>` sets the five degrees of freedom in RAW mode.

8. Configurations (Figs. 1, 11)

The same skeleton, fingers and actuators are used in two configurations: a robotic gripper with a flange and alignment pin for mounting on a robot wrist, powered from the robot tool connector; and a prosthetic hand with a forearm cover, socket cover and a wrist joint with soft bearings.

Figures

Sources: `hardware/renders/` and `docs/Assembly Instructions.pdf` of the repository; Figs. 8–10 drawn for this disclosure from the published firmware and assembly instructions.



Fig. 1 — Assembled hand, front and back renders.

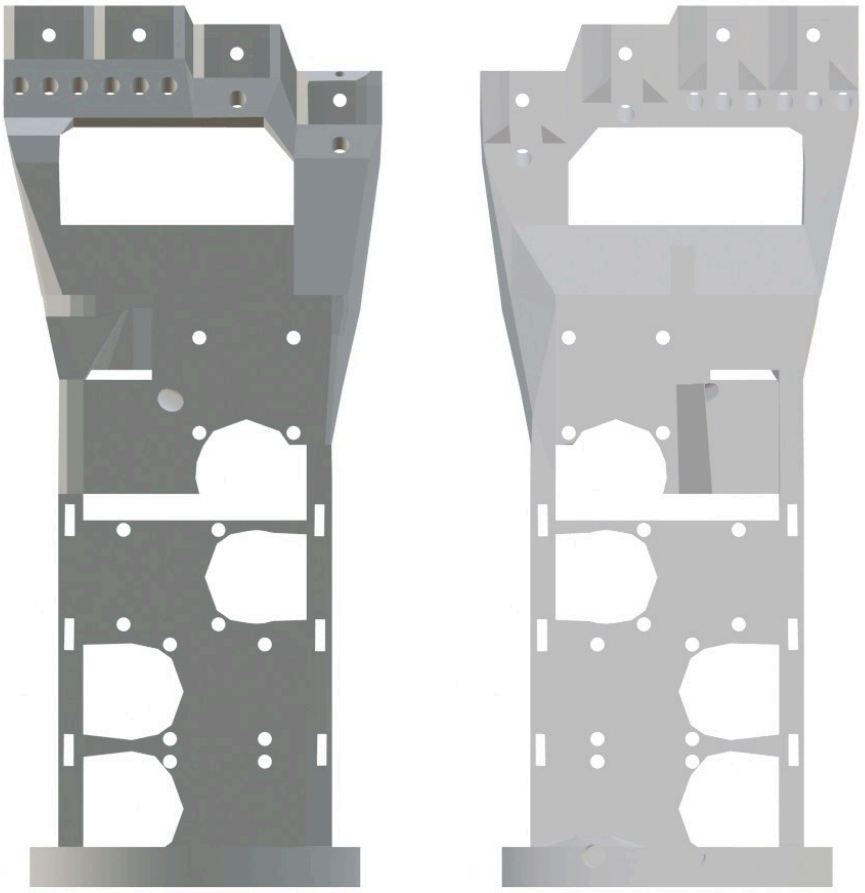


Fig. 2 — Internal skeleton, front and rear views.

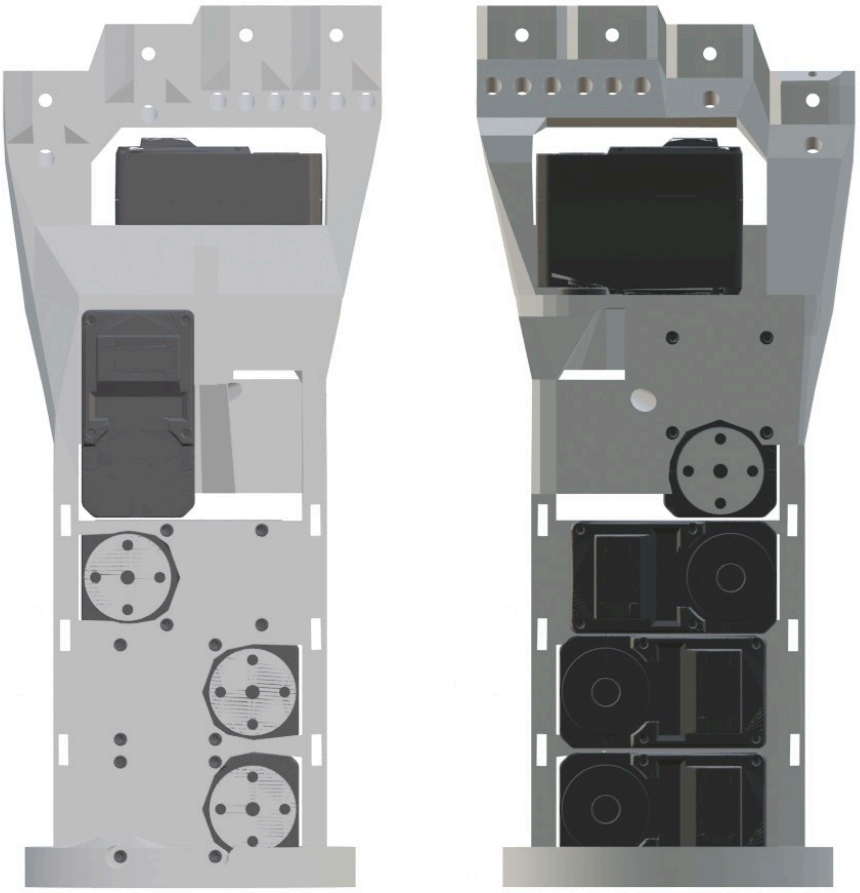


Fig. 3 — Skeleton with the five actuators installed, front and rear views.

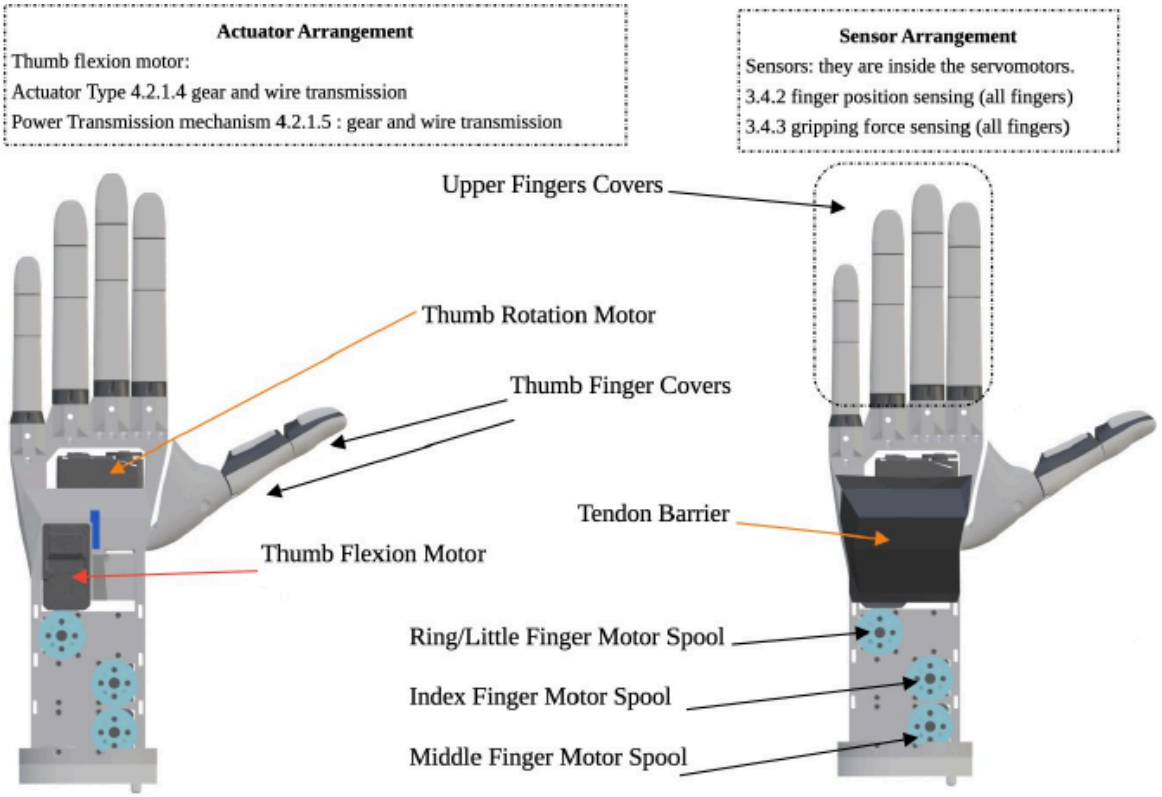


Figure 18: Left Hand - Arrangement of actuators and sensors

Fig. 4 — Arrangement of actuators, spools and tendons-barrier.

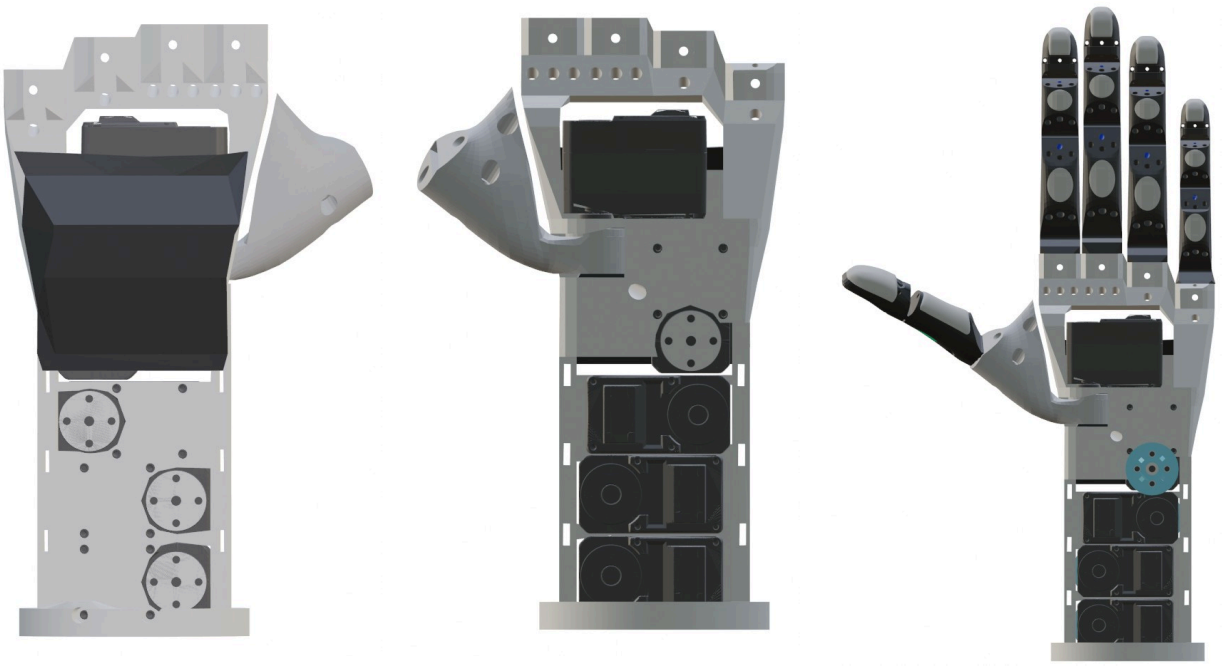


Fig. 5 — Thumb base mounted on the thumb rotation actuator flange (left, front and rear); thumb installed (right).

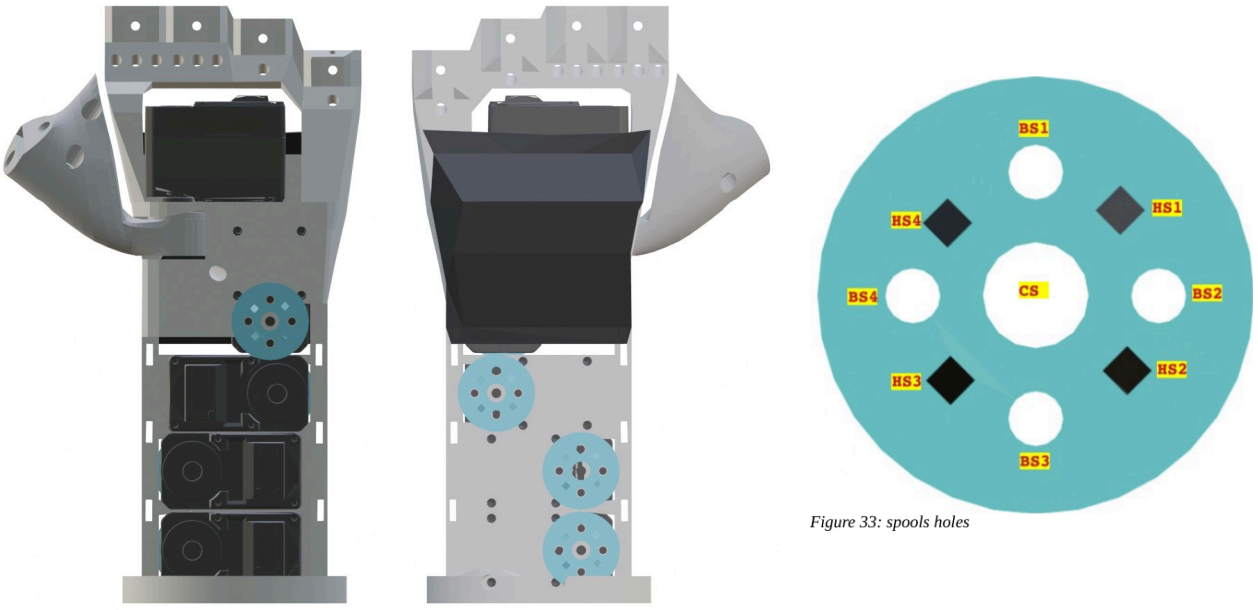


Figure 33: spools holes

Fig. 6 — Spools on the actuator flanges (left); spool holes (right): HS1–HS4 tendon holes, BS1–BS4 locking bolts, CS centre.

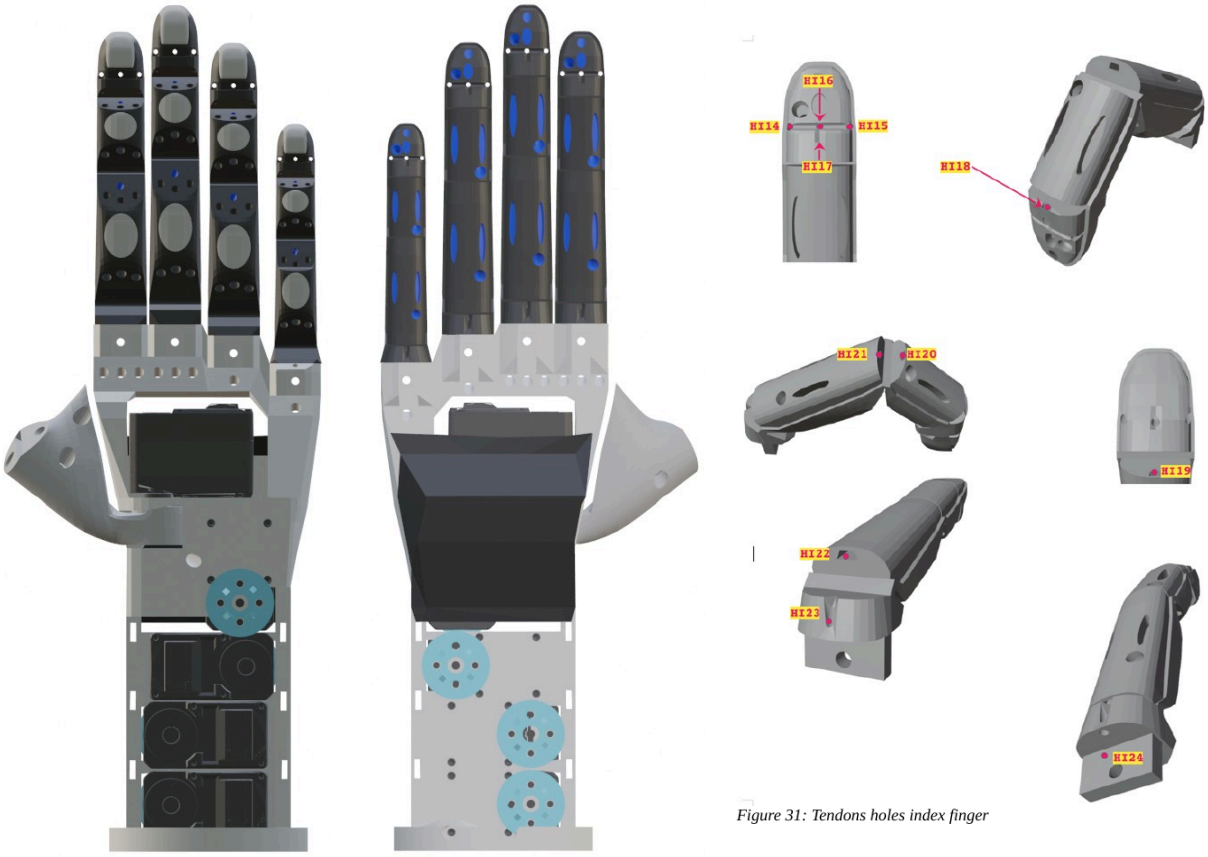


Fig. 7 — Upper fingers installed on the skeleton (left); tendon holes of the index finger (right).

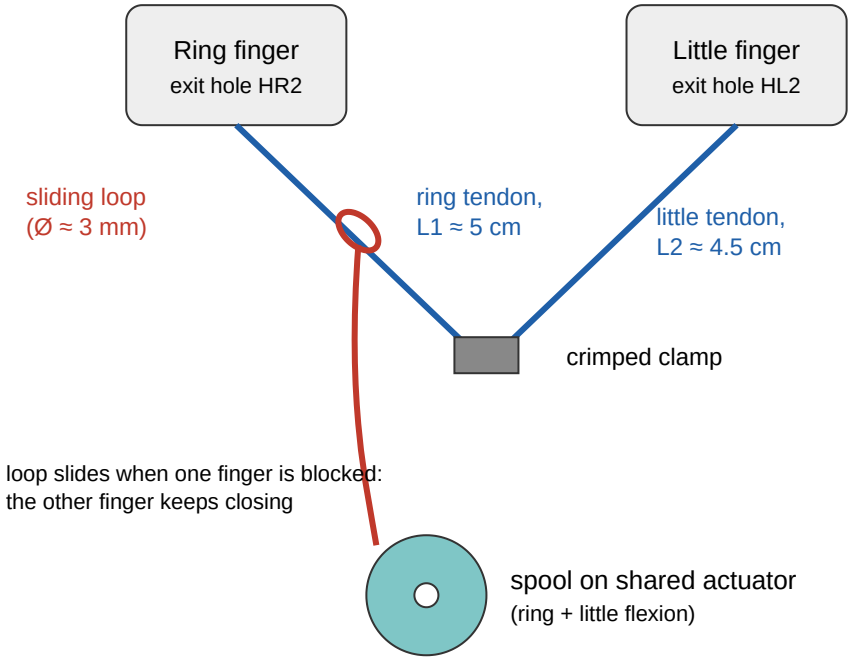


Fig. 8 — Schematic of the ring–little coupling: crimped clamp joining the two finger tendons and sliding loop pulled by the shared actuator spool.

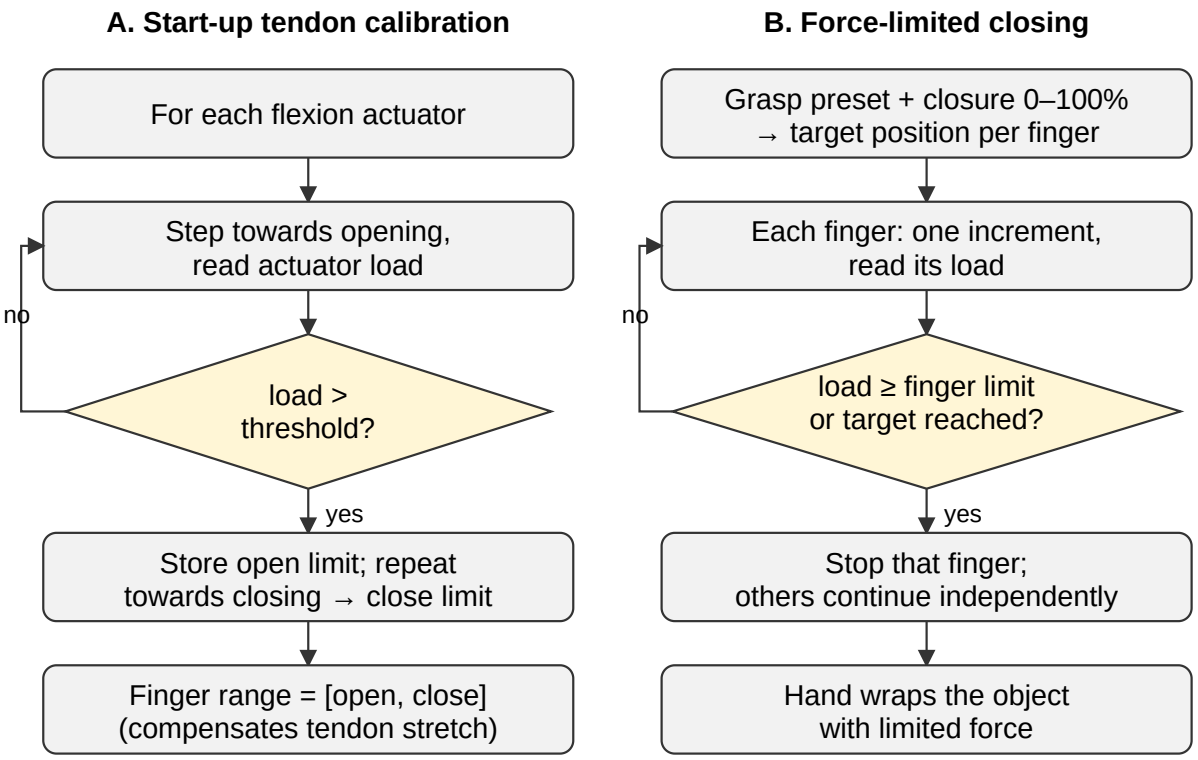
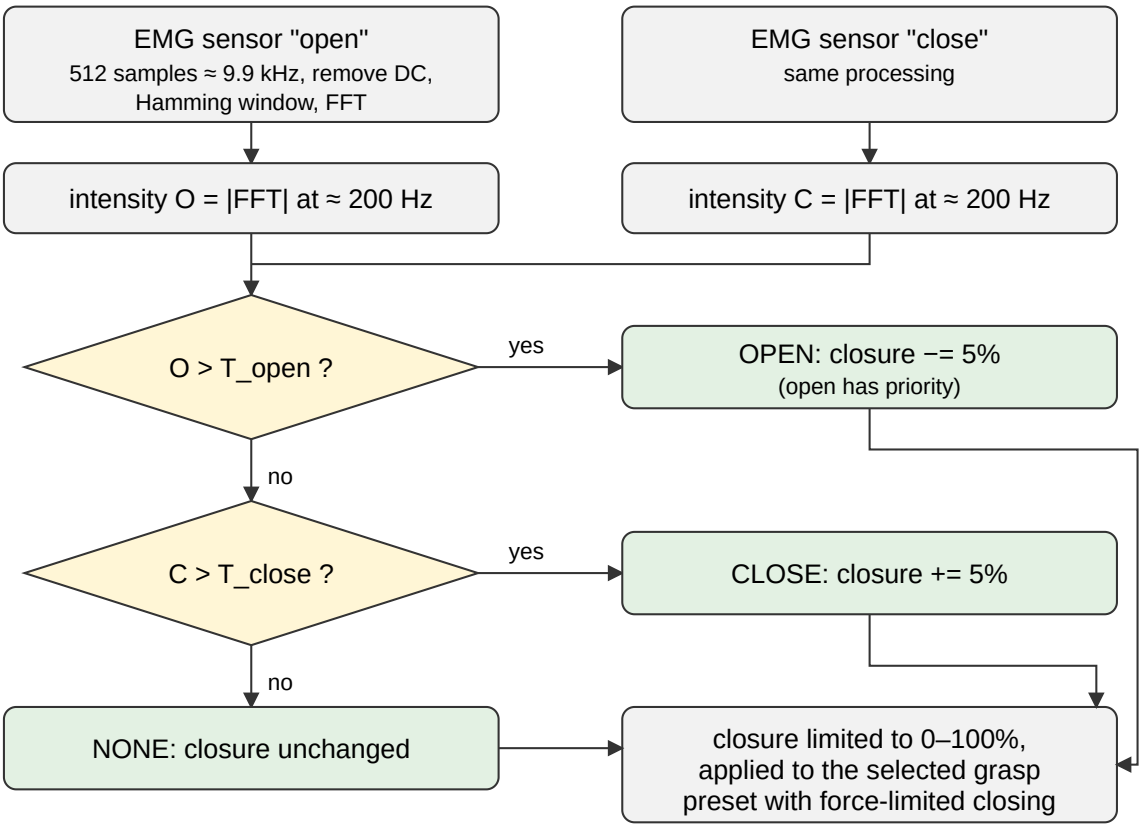


Fig. 9 — Flow chart of start-up tendon calibration (A) and force-limited closing (B).



Each channel has its own threshold (T_open, T_close). If closure stays at 0% for 60 s the hand goes to the relax posti

Fig. 10 — Flow chart of the two-channel EMG decision rule (prosthetic version).

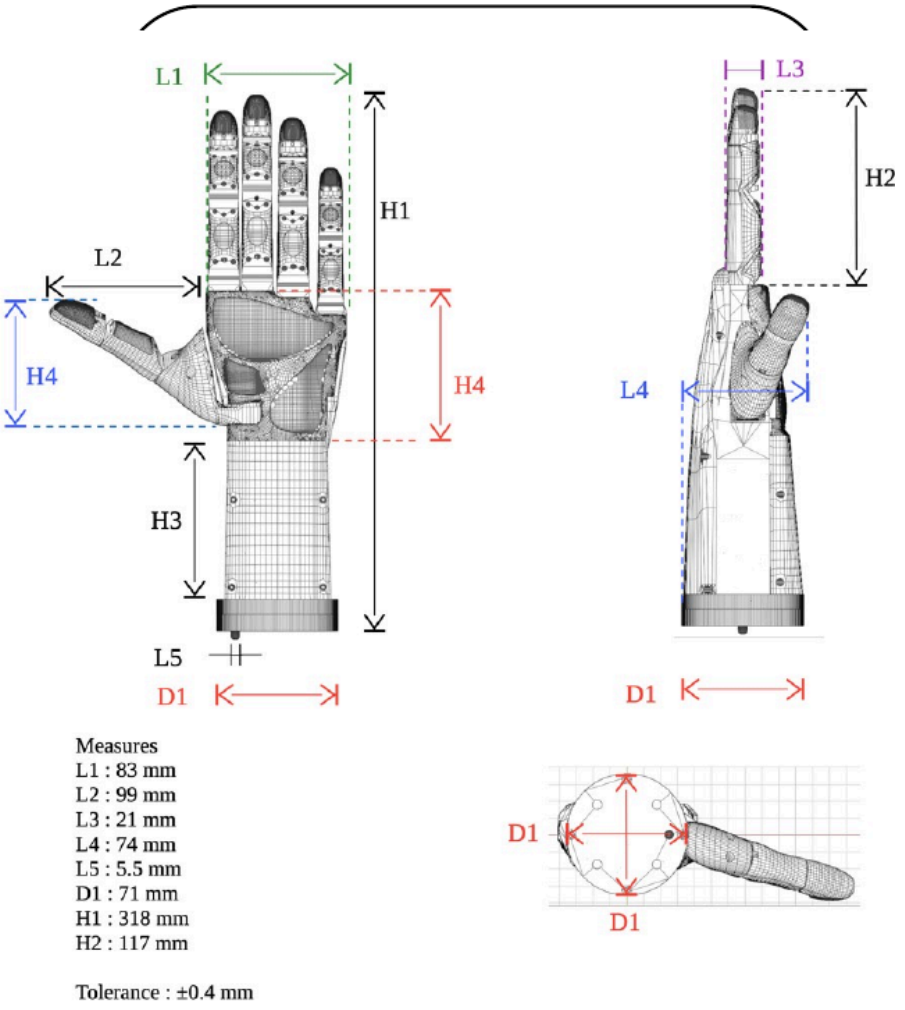


Figure 19: Left Hand Measures

Fig. 11 — Main dimensions of the hand (L1 83 mm, L2 99 mm, L3 21 mm, L4 74 mm, L5 5.5 mm, D1 71 mm, H1 318 mm, H2 117 mm; tolerance ±0.4 mm).

Alternative embodiments

The actuators may be any rotary actuators with position and load or current sensing (smart servos, DC or BLDC motors with encoders and current measurement). The number of fingers, phalanges and actuators may vary from one to six actuators; more fingers may share an actuator through a cascade of sliding loops, levers or pulleys. Tendons may be made of nylon, UHMWPE, aramid or steel cable and may run in PTFE tubes, sheaths or over pulleys. Flexion and extension tendons may use spools of different or non-circular profile. The skeleton may be printed, moulded, cast or machined. Load sensing may be replaced or complemented by fingertip force sensors, tendon tension sensors or tactile skins. Commands may come from a robot controller, a computer, a smartphone, EMG or other biosignal sensors, or a camera-based grasp selector. EMG intensity may be computed as band power over several FFT bins, RMS, envelope or wavelet features instead of a single bin; one, two or more EMG channels may be used, with proportional instead of fixed-step closure changes, adaptive or user-calibrated thresholds, or pattern-recognition classifiers. The grasp may be selected by manual positioning of any back-drivable joint, not only the thumb. Further variants are listed in `VARIANTS.md` of the repository, which is part of this disclosure.

Advantages

- Compact: all actuators inside the hand skeleton, no forearm required.
- Active opening and closing with one actuator per finger, no return springs.
- Adaptive grasp of irregular objects with fewer actuators.
- Automatic calibration compensates tendon stretch and assembly tolerances.
- Limited, controllable grasp force for fragile objects and safe interaction.
- Low cost: printed parts and off-the-shelf components.
- One design for both robotic and prosthetic applications.

Applications

Robot end effectors, collaborative robots, humanoid robots, teleoperation, upper-limb and partial-hand prostheses, research and education, rehabilitation and exoskeleton devices, remote handling.